

YEAR OF STUDY: TWO

DATE: TUESDAY, 2ND DECEMBER 2014

SEMESTER: ONE

TIME: 2:00 PM-5:00 PM

INSTRUCTIONS:

1. Attempt any four (4) questions

Question one

- Who is a systems analyst? (3 marks)
- Differentiate between systems analysis and systems design (4 marks)
- Discuss the qualities of a systems analyst (18 marks)

Question two

- Define the term requirements analysis (4 marks)
- Identify the fact finding methods as used in requirements analysis highlighting advantages and disadvantages of each (21 marks)

Question three

Prototyping is one of the most recommended system development approaches for use by system designers. Using your knowledge in systems development, answer the following questions

- Differentiate between prototyping and throwaway prototyping (5 marks)
- Identify the steps undertaken when developing systems using the prototyping approach (10 marks)
- Why would you prefer to use prototyping approach as opposed to other approaches in system development? (10 marks)

Question four

- a) Cryptography has been used for many years to ensure that information is protected. Explain how asymmetric algorithm can be used to protect information which is transmitted. (5 marks)
- b) Discuss the three (3) principles or goals of information system security together with possible threats that compromise them, and the countermeasures that can be adopted to address them. (12 marks)
- c) Security controls are necessary to address security threats in an organization. Discuss three categories of security controls that can be used in organizations. (8 marks)

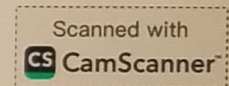
Question five

- a) Establishing a system boundary is very important in system design and development. With illustration, explain your understanding of a system boundary (5 marks)
- b) Using symbols of a structured approach to system analysis and design, illustrate the relationship between the following entities. (10 marks)
 - i. Vehicle: Part
 - ii. Doctor: Patient
 - iii. Course: Student
 - iv. Student: School bag
 - v. Faculty: Department
- c) Given 3 entities of student, library and lecturer in Makerere University Business School. Illustrate a conceptual, logical and physical data models based on the above entities. (10 marks)

Question six

- a) The Truman's triangle is a good guide in network design. Using an illustration, explain how this triangle is used by Network designers. (5 marks)
- b) Using examples where applicable, explain the following concepts which are applied in database development (2 marks each)

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QUESTION FIVE

Focusing on an organization of your choice as a case study, use relevant examples to discuss the following.

- a) The difference between formal and informal meetings held in that organization. (2 Marks)
- b) The four types of formal meetings held in the organization. (8 Marks)
- c) The roles of the Chairperson in one of the meetings held by the organization. (15 Marks)

END

JK